# SaiDeepak Gupta Karnatakapu

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## SUMMARY

Software engineer with **3+ years** of professional experience building production services and **LLM applications** in Python and Java. Ships AI with **evaluation, guardrails, and monitoring** in privacy-sensitive, regulated environments, from KYC on Citi's 48M-user banking platform to generative AI at Rokt. Experienced in object-oriented design, test-driven development, Agile delivery, and on-call operations on AWS.

## TECHNICAL SKILLS

**Generative AI:** LLM evaluation & guardrails, structured outputs, hybrid RAG (vector + BM25, Reciprocal Rank Fusion), agentic workflows (LangGraph, MCP), on-prem inference, LangChain, Claude, OpenAI, Pinecone, ChromaDB

**Full-Stack & Event-Driven**: Java, Spring Boot, Python, FastAPI, Kafka, REST microservices, idempotent event processing, Redis caching, React, Next.js, TypeScript

**Data & Cloud:** PostgreSQL (row-level security), MySQL, SQL, AWS (EKS, EC2, S3, RDS, API Gateway, Elastic Beanstalk), Kubernetes, Docker, Terraform

**Reliability & Compliance**: OpenTelemetry, Prometheus, Jenkins CI/CD, TDD (JUnit, pytest), OAuth/JWT, GDPR, CCPA, FERPA, KYC

## EXPERIENCE

**Rokt** (Contract via Axitem) **Aug 2025 – Present**

*Software Engineer* *New York*

*Customer data platform (mParticle CDP) | Python, FastAPI, Java, React/TypeScript, Redis, OpenTelemetry, AWS*

- Built a **React/TypeScript** identity-tracing tool for the mParticle CDP, using **React Query** to fetch and cache data from **Java/Spring Boot APIs** to visualize profile merges, cutting investigation time by about **30%**.
- Reduced **p95 latency** of real-time decisioning APIs by **29% (185 ms to 132 ms)** under production load with Redis caching and parallel **asyncio** retrieval.
- Partnered with product managers and data scientists to integrate identity-resolved and predictive customer attributes through Java/Spring Boot APIs and event-driven workflows, enforcing GDPR consent and CCPA opt-out before data reached decisioning.
- Improved production reliability with **OpenTelemetry** tracing, **Prometheus** metrics, and **Chronosphere** alerting as part of the on-call rotation, helping the team cut incident detection time from 20 to 8 minutes and resolution time from 60 to 25 minutes.
- Engineered **Python LLM evaluation** and **guardrail workflows** for Rokt's generative ad-content engine, using structured outputs and automated quality checks to reduce manual rework of generated headlines and calls-to-action.
- Wrote **Python and Java services test-first (TDD)** with **pytest and JUnit** at **85%** coverage; contributed to design and code reviews in Agile sprints.

**Binghamton University** **Jan 2025 – May 2025**

*Software Engineer (Part-time, during MS)* *Binghamton, NY*

- **Owned an on-premises AI teaching agent** serving **~200 student queries/day** with zero cloud APIs, keeping student records on-premises in line with FERPA.
- Improved answer accuracy with a **hybrid RAG pipeline** (LangChain, ChromaDB, Ollama) that merges vector and BM25 search through Reciprocal Rank Fusion.
- Reduced context overhead by exposing syllabus, lab, and concept docs as **3 MCP tools**, orchestrated by a **LangGraph agent** with step limits and fallbacks that prevent loops and silent failures.
- **Cut inference latency 43% (1.2 s to 680 ms) by serving Llama 3.1 8B with 4-bit quantization and asyncio request batching.**
- **Saved ~$400/month** versus cloud LLM APIs through fully on-premises inference, backed by a cost-per-query analysis.

**Virtusa** (Client: Citi) **May 2021 – Jul 2023**

*Software Engineer* *Hyderabad, India*

- Built **distributed Java/Spring Boot services** and React features for customer onboarding and identity verification (KYC) on a **48M-user banking platform**, delivering across the full SDLC from requirements to release.
- Integrated Onfido identity verification with status updates streamed through **Kafka**, **cutting verification time by about two-thirds**, and autoscaled Kafka consumers on **AWS EKS** for peak event volume.
- Integrated **15+ third-party REST APIs** into the payments and banking module; provisioned AWS (EC2, S3) with Terraform.
- Automated **Jenkins CI/CD** build pipelines within Scrum sprint delivery, **reducing post-deployment issues by 30%**.

**Virtusa** **Nov 2020 – Mar 2021**

*Software Engineer Intern* *Hyderabad, India*

- Built a reusable **Spring Boot/MVC REST API layer for data retrieval**, significantly reducing the time developers needed to add new endpoints.
- Built an **LDAP/JWT authentication system** with **90% JUnit unit test coverage**.

## EDUCATION

**State University of New York at Binghamton,** *Master of Science in Computer Science* *Aug 2023 – May 2025*

***Relevant Coursework:*** Operating Systems, Distributed Systems, Database Systems, Data Mining, Design Patterns

**Malla Reddy College of Engineering and Technology,** *Bachelor of Technology in Information Technology*

## PROJECTS

**Source-Grounded RAG Chatbot (Freelance, NDA)** | Python, LangChain, Pinecone, FastAPI, AWS

- Built a **RAG chatbot** that answers questions from a website's own content, chunking and embedding pages into Pinecone, served through FastAPI on **AWS Elastic Beanstalk with auto-scaling** and JSON schema-validated responses.
- **Cut hallucination rate from 35% to under 8%** by enforcing source-grounded answers, each linked to the exact document chunk it came from.

**HireFast – Multi-Tenant Candidate Search** | TypeScript, Next.js, PostgreSQL, Stripe, Claude API

- Designed the **PostgreSQL schema and PL/pgSQL search over 5,000 profiles**, using the **Claude API** to turn free-text recruiter queries into GIN-indexed lookups with index-range result paging.
- Isolated tenant data with **row-level security** under deny-by-default grants, and made the free-tier limit **race-proof with one atomic conditional UPDATE**, verified by automated concurrency tests (10 requests: 1 success, 9 rejected).
- Built a **Stripe webhook handler** with HMAC verification, idempotent processing keyed on event ID, and an out-of-order event guard.

**EMG Muscle-Signal Classification Study |** Python, NumPy, scikit-learn

- Implemented logistic regression (gradient descent, L2 regularization, early stopping) and k-means (Hungarian label matching) from scratch, and showed both perform near chance (~50–58% accuracy) on balanced EMG muscle-signal data, indicating non-linear structure that linear and centroid-based models cannot capture.

## CERTIFICATIONS

AWS Certified AI Practitioner | AWS Certified Cloud Practitioner